

Metro smart parking node flow design

Node-RED is triggered using the command ‘node-red’. Through the URL <http://127.0.0.1:1880/>, the node-RED UI flow browser is enabled. We have considered that the Node-RED setup is done and working.

Follow the steps given below to create the login and submission forms.

Login form

- 1) From the node palette, drag and drop *http* in node, which creates an HTTP end point for creating Web services.
- 2) Connect *http* in node to *function* node. The latter helps

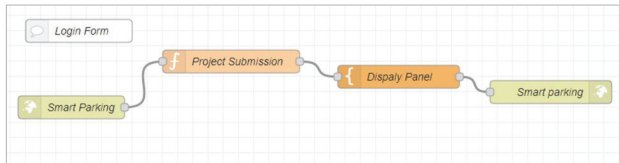


Figure 4: Login form for smart parking for cars

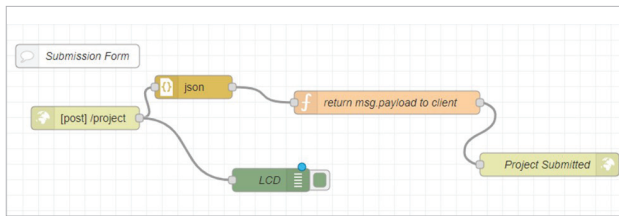


Figure 5: Submission form for smart parking for cars

Edit http in node

Delete

Cancel

Done

⚙ Properties

⚙

📄

🔗

Method

GET

URL

/MetroStation

Name

Smart Parking

☐ Enabled

Figure 6: *Http in* node property configurations

Edit function node

Delete

Cancel

Done

⚙ Properties

⚙

📄

🔗

📌 Name

Project Submission

📌

Setup

Function

Close

1 msg.url = "project";

2 return msg;

🔗 Outputs

1

⬆

⬆

🔗 Enabled

Figure 7: *Function* node property configurations

to code JavaScript functions to run against the messages being received by the node.

- 3) Connect *function* node to *template* node, where the latter creates a Web API based on the provided template.
- 4) Connect *template* node to *http response* node; the latter sends responses back to requests received from an *http input* node.

Submission form

- 1) Drag and drop *http* in node and connect it to *json* node, which converts and communicates the data as JSON string.
- 2) Connect *http* in node to *debug* node, which gives output

Cancel

Done

Properties

Name

Dispay Panel

Property

msg. payload

Template

Syntax Highlight: HTML

1

<!DOCTYPE html>

2

<html>

3

4

<head>

5

<h1 style="background-color: pink; ALIGN="right">

6

7

</head>

8

<div style="background-image: url(https://www.wu

9

<body style="background-color: lightBlue; ">

10

<IMG SRC="https://www.mindgames.com/uploader

11

</> Format

Mustache template

→ Output as

Plain text

☐ Enabled

Figure 8: *Template* node property configurations

Edit http response node

Cancel Done

Properties

Name Smart parking

Status code msg.statusCode

Headers

+ add

The messages sent to this node **must** originate from an *http input* node

Enabled

Figure 9: *Http response* node property configurations